

Akella Raghavendra Sir
Founder & Chief Mentor

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Nerve Agents

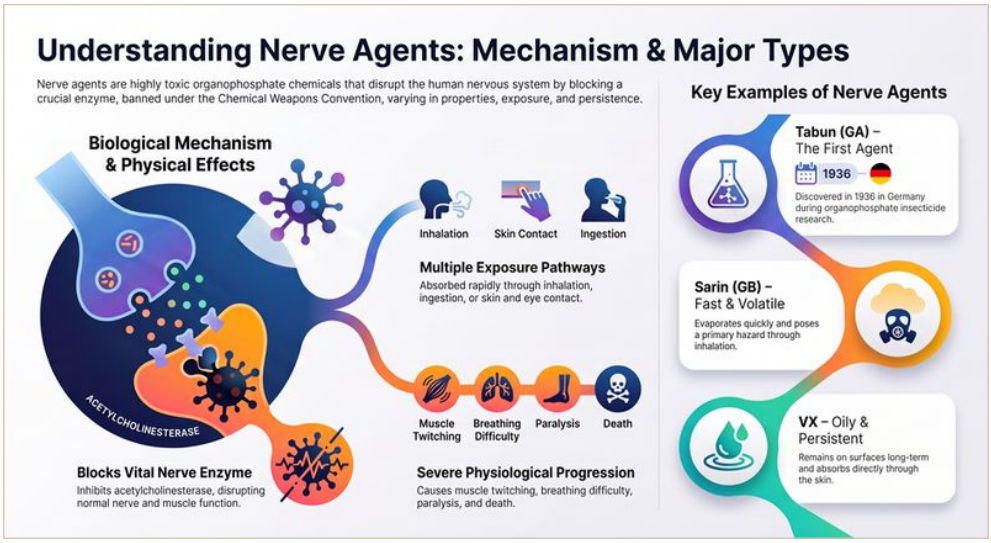


Nerve agents are highly toxic chemical warfare agents that disrupt the nervous system by interfering with acetylcholine regulation. The article examines their mechanism, major types, health effects, international controls and challenges in preventing chemical-weapons misuse.

Nerve Agents

Syllabus areas:

GS Paper III – Science & Technology, Internal Security, Disaster Management



Nerve agent is a highly toxic chemical that disrupts nervous system in the human body by **blocking acetylcholinesterase**, an enzyme needed for normal nerve and muscle function.

Nerve agents can be absorbed by **inhalation, ingestion** or contact with skin or eyes.

Nerve agents block acetylcholinesterase, an enzyme needed for **nerve signalling, causing muscle twitching, breathing difficulty, paralysis and death**.

Examples of major nerve agents:

- Tabun (GA)

- Sarin (GB)
- Soman (GD)
- VX

Nerve agents are first time discovered by German Scientists while **developing insecticides**. In **1930s**, German chemists studying organo phosphate insecticides discovered compounds such **as tabun and sarin** that were extremely toxic to nervous system.

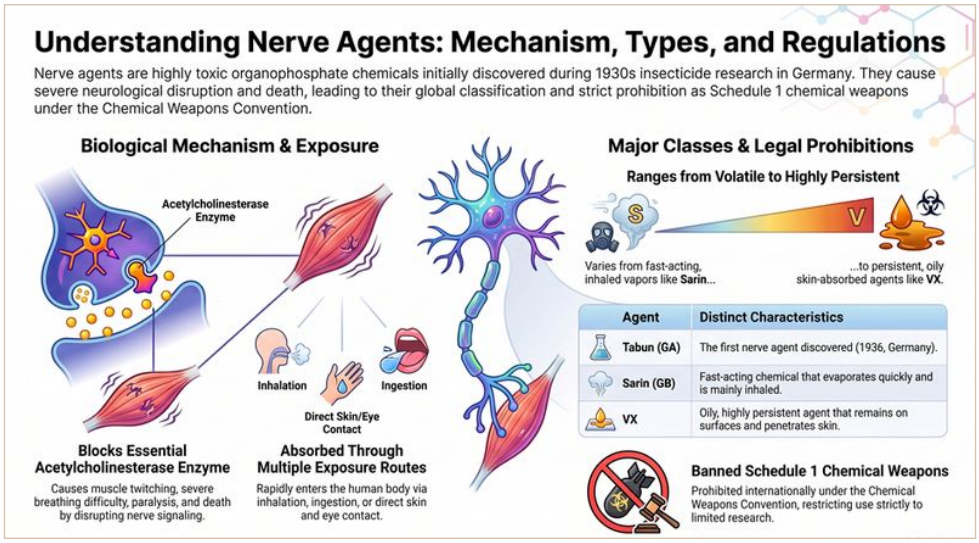
therefore Tabun – is the first nerve agent discovered in 1936 in Germany.

Sarin – A fast acting nerve agent that evaporates quickly and is mainly inhaled.

Soman – More persistent than sarin and harder to treat after exposure.

VX – An oily nerve agent that can remain on surfaces for long periods and be absorbed through skin.

A-series agents such as A-232, A-234 which are commonly discussed as **Novichok agents.**



All these nerve agents are prohibited chemical weapons under Chemical Weapons Convention. These are classified as **Schedule 1 chemicals**, meaning their permitted uses are extremely limited to purposes such as research, medical, pharma, agriculture under Controls.

There are also other chemical warfare agents such **as mustard agents** and certain **choking agents** but these belong to different chemical classes and act differently from nerve agents.

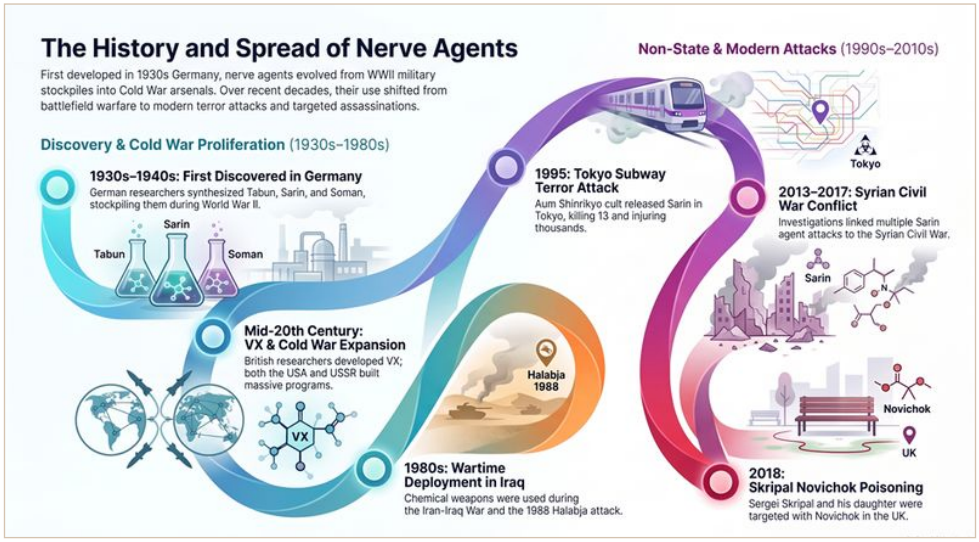
Chemical warfare means using toxic chemicals as weapons to kill, injure, incapacitate, or otherwise harm people or animals during armed conflict.

Examples include:

- **Nerve agents:** Sarin, tabun, Soman, VX.
- **Blister agents:** Sulfur mustard
- **Choking agents:** Chlorine, phosgene
- **Blood agents:** Hydrogen cyanide

Chemical Weapons Convention prohibits development, production, stockpiling, transfer and use of chemical weapons.

Brief History of Nerve agents

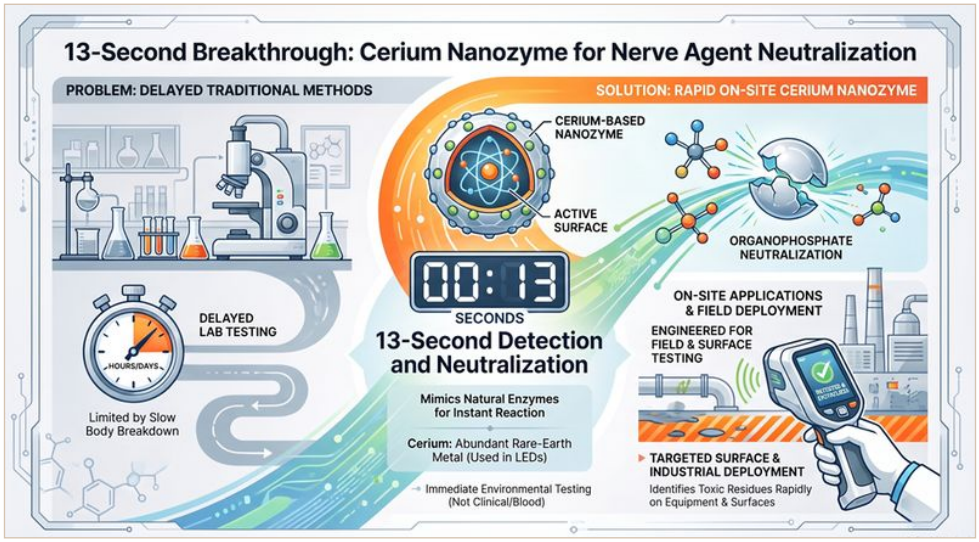


- **Germany – 1930s –** the first known nerve agent, tabun was discovered (1936) Sarin (1938) Soman (1944)
- **World War II –** Germany produced and stockpiled nerve agents, but they were not used in battlefield.
- **British Researchers** developed the compound that is VX.
- **Cold War –** USA, USSR maintained substantial chemical weapons programs.
- **Halabja chemical attack** in Iraq in 1988, killed thousands.
- **Iran-Iraq war (1980's) –** Iraqi forces used chemical weapons.
- **Tokyo subway attack – 1995 –** Members of Aum Shinrikyo cult released sarin gas in the Tokyo subway, killing 13 people and injuring thousands.
- **Syria (2013-2017) –** investigations linked multiple attacks involving sarin to Syrian Civil War
- **Sergei Skripal Poisoning – 2018 –** Former Russian spy Sergei Skripal and his daughter were poisoned in UK with a Novichok nerve agent.

Detecting Nerve agents in the Human Body

Nerve agents can be detected in the body, but they are often broken down rapidly, so testing is time-sensitive. Blood tests, urine tests, environmental sample tests etc are available.

Now Indian scientists have found a solution: **Nano-material** that can detect and neutralise nerve agents within 13 seconds.



They developed cerium based **nanozyme**, a material engineered to behave like a natural enzyme, that reacts quickly with **organophosphate compounds** (nerve agents).

Cerium is the most abundant rare-earth metal and helps create the bright white light in energy saving bulbs and LEDs.

Interesting point is that – this nanozyme is designed for on-site detection of contamination, not for medical or blood testing. It can be used to identify toxic residues on surfaces, equipment, or in industrial and environmental settings – situation where rapid, reliable detection is critical.